

Title

Minor Project Report

*submitted in partial fulfillment of the
requirements for the award of the degree of*

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & TECHNOLOGY

by

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Under the guidance of
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School of Computer Science, UPES
Bidholi, Via Prem Nagar, Dehradun, Uttarakhand
Month – 2025

CANDIDATE'S DECLARATION

I/We hereby certify that the project work entitled “**Title of Project**” in partial fulfillment of the requirements for the award of the Degree of **BACHELOR OF TECHNOLOGY** in COMPUTER SCIENCE AND ENGINEERING with specialization in and submitted to the Data Science Cluster, School of Computer Science, UPES, Dehradun, is an authentic record of my/our work carried out during a period from **Month, Year** to **Month, Year** under the supervision of **Guide Name(s), Designation and Affiliation**.

The matter presented in this project has not been submitted by me/us for the award of any other degree of this or any other University.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Date: _____ 2025

Name of Guide
Project Guide

Acknowledgement

We wish to express our deep gratitude to our guide **Name**, for all advice, encouragement and constant support he/she has given us throughout our project work. This work would not have been possible without his support and valuable suggestions.

We sincerely thanks **Name** our respected Cluster Head, Data Science, School of Computer Science, for his great support in doing our project _____. We are also grateful to Dean SoCS UPES for giving us the necessary facilities to carry out our project work successfully. We also thanks to our Course Coordinator (NAME) for providing timely support and information during the completion of this project.

We would like to thank all our friends for their help and constructive criticism during our project work. Finally, we have no words to express our sincere gratitude to our parents who have shown us this world and for every support they have given us.

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Abstract

“TSAR” – The Target Seeking Autonomous Robot is a project which is going to deal with the complexity of collision avoidance, computational processing, localization, environmental sensing and target seeking. An autonomous robot is a self-piloted machine that does not require a human intervention to accomplish its tasks. These robots are designed to perform desired tasks in unstructured environments without continuous human guidance. The goal of the project is to design and build a low cost autonomous vehicle control system for a ground vehicle. The vehicle must be able to navigate through the predetermined route (desired destination) and avoid obstacles while maintaining an optimum speed and also restoration of the original path to reach its predefined destination successfully. The demonstrator system uses BOE-BOT. The bots are initially placed at some position (which is referred as Home Position hereafter) and a target (Destination Address hereafter) is given to them. Then the required processing is done and the predefined path is analyzed and fixed for locomotion. The bots are programmed such that they can detect as well as avoid all the objects in their path and no collision occurs. Collision avoidance is achieved by using infrared sensors. Once the Bot goes out of the predefined path the Path Restoration Technique comes into action in order to reach its original destination. One possible task of an autonomous vehicle is to navigate a pre-programmed route while avoiding any obstacles the vehicle may encounter. This function is useful in applications such as

- Surveillance Robot,
- Safety System,
- House hold robots
- Autonomous Moving Robot
 - In the Air
 - On road
 - Underwater

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5.1 Path Tracing using a Cartesian Coordinate System

5.2 Scenarios

5.3 Algorithms

5.3.1 Scenario 1: Coordinate-Based Seeking

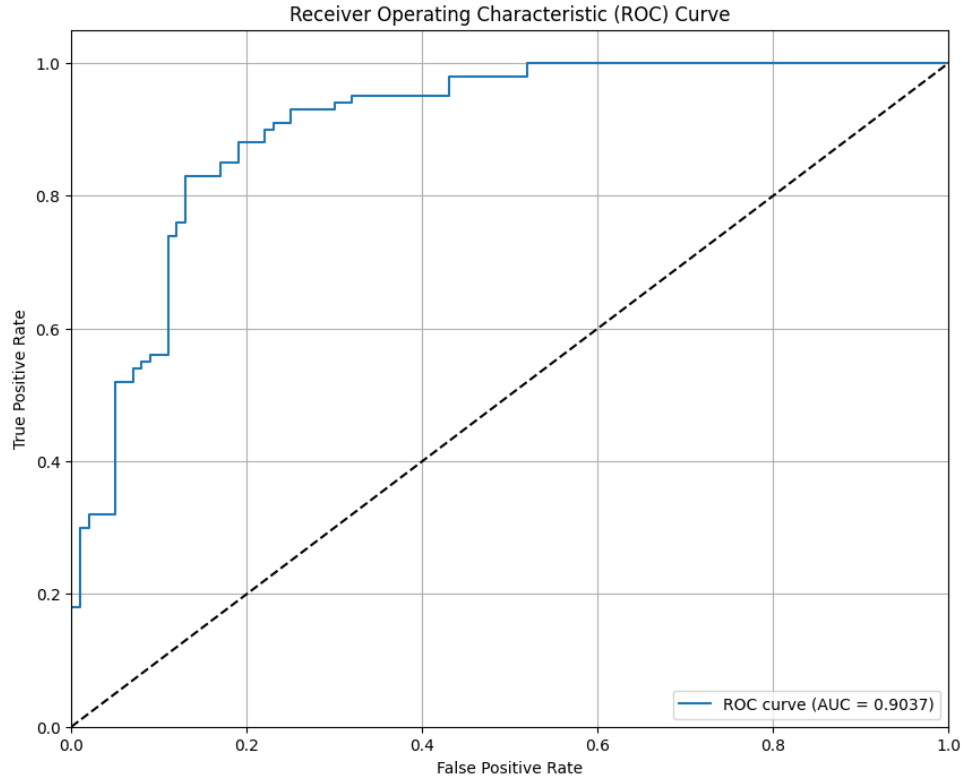


Figure 5.1: Receiver Operating Characteristic (ROC) Curve showing model performance with $AUC = 0.9037$.

5.3.2 Scenario 2: Lookup-Based Seeking

5.3.3 Scenario 3: Serpentine Scanning

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Appendix A

BASIC Stamp and Carrier Board Components and Features

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References

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Bibliography

- [1] Reference as per appearance in the chapters.